



The Jing-Zhang AI innovation belt is not a city that "uses AI to assist planning". It is a city whose governance is designed as an Agent OS — participation, approval, piloting, evaluation and knowledge capture are all turned into protocols, driven by events and fully auditable. AI and citizens collaborate on one shared protocol, and humans keep the final decision.



Programmable City

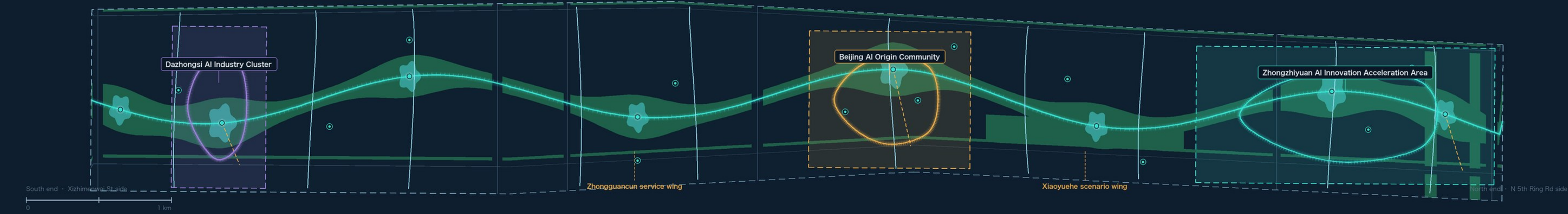
Jing-Zhang AI Belt urban governance protocol design | Overall spatial structure: layers (four-layer protocol section) + loops (one spine · three loops · eight protocol nodes)

FIG 01 / SITE OVERVIEW

PROVISIONAL · not an official redline

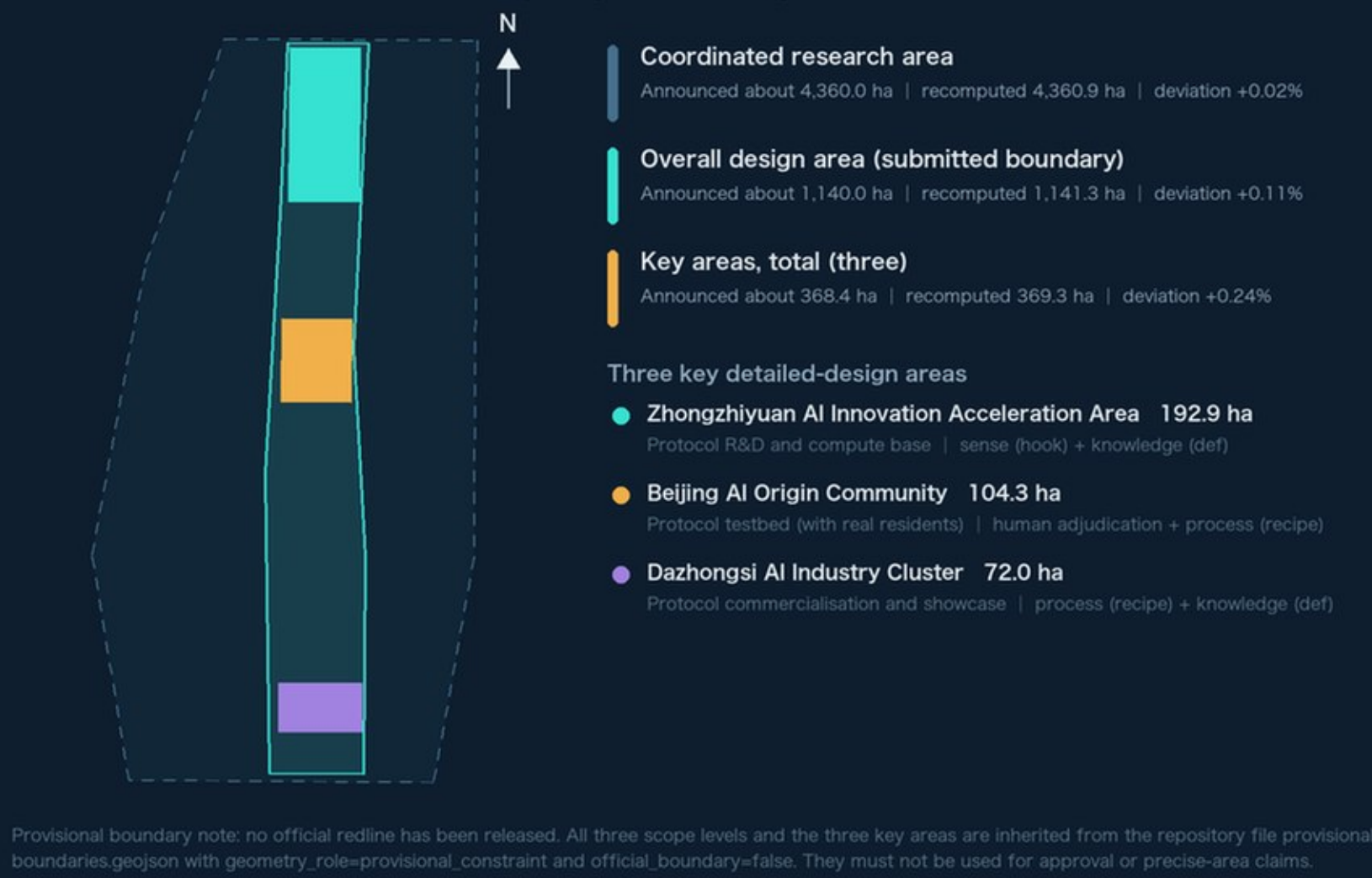
Spatial structure of the overall design area (about 11.41 sq km)

The Jing-Zhang heritage alignment becomes a longitudinal "protocol spine", each key area closes into its own "protocol loop". Spine = protocol bus, loop = a governance cycle that keeps running



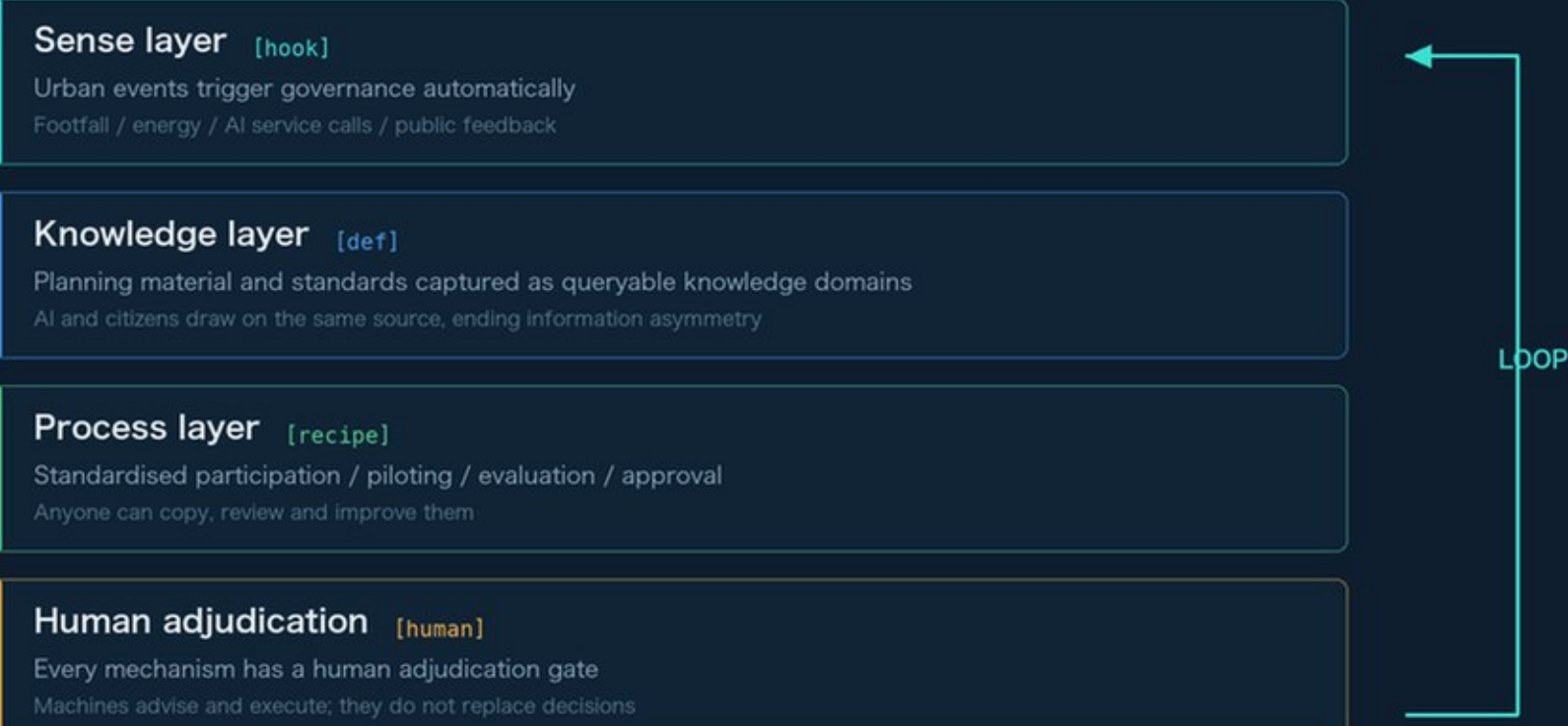
Three scope levels and their nesting

All areas recomputed in EPSG:4548 and checked line by line against announced figures



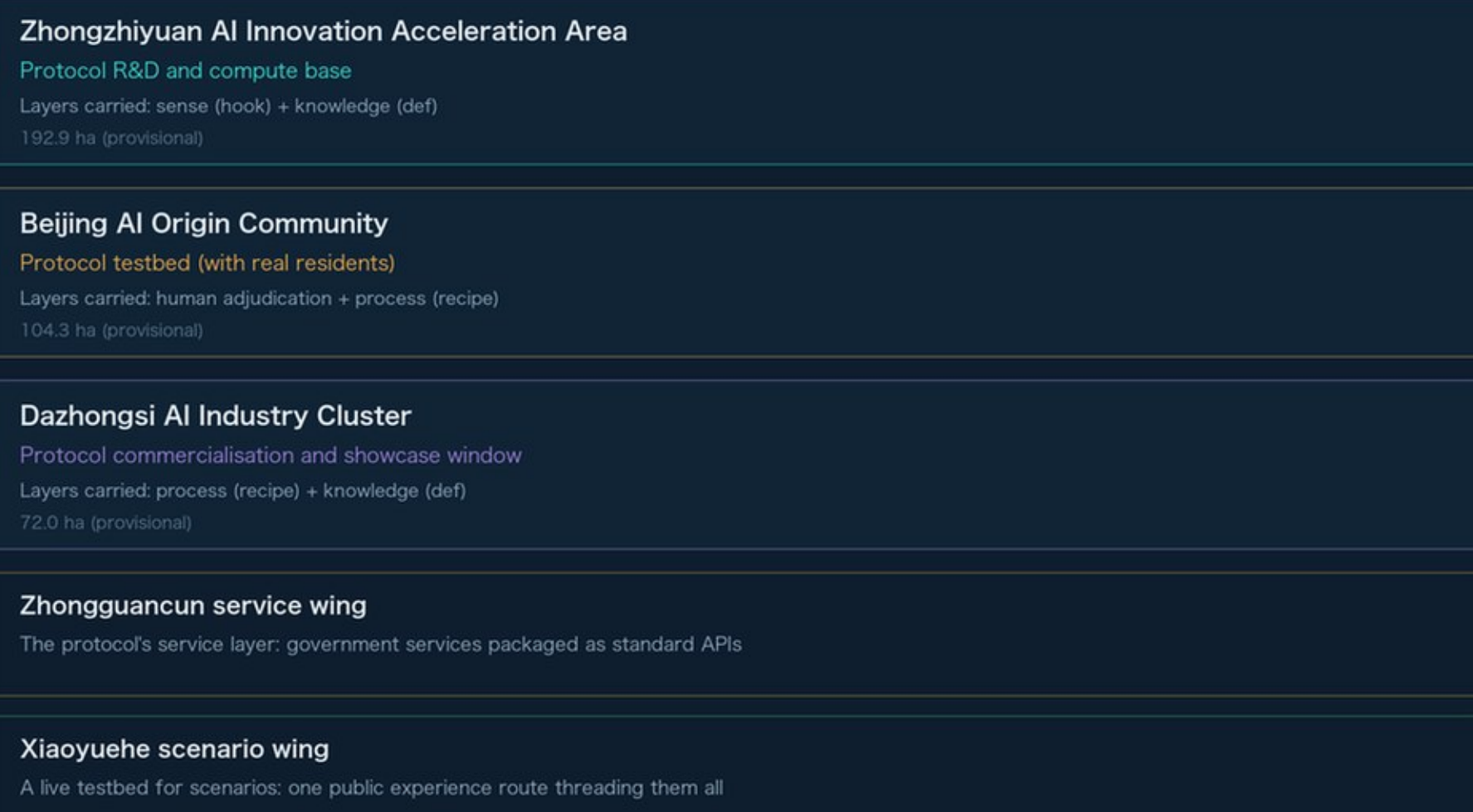
Four-layer city OS architecture (layering)

frago's hook / def / recipe / human final judgement, mapped layer by layer onto urban governance



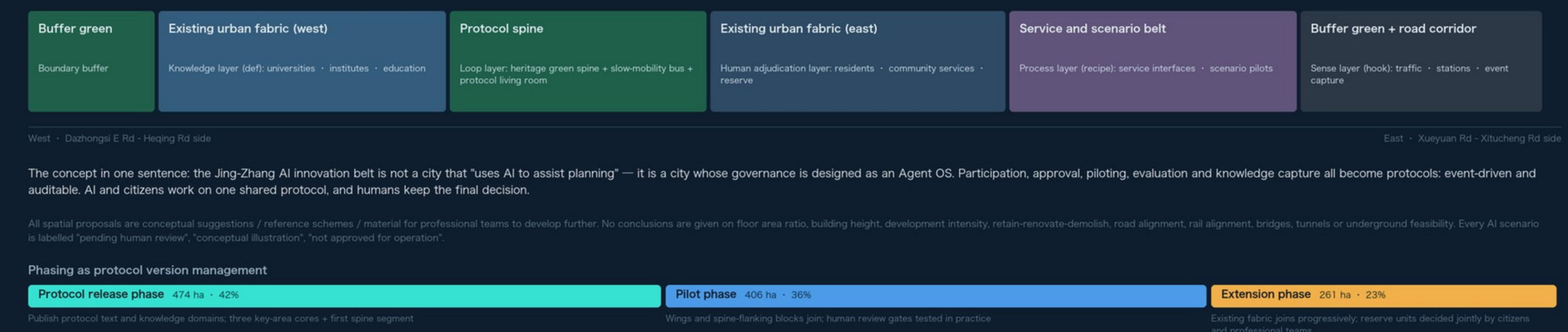
Roles of the three areas and two wings under the protocol

Not conventional functional zoning — the same protocol playing different roles at different stages



Corridor section: four protocol layers on one cut (layering)

Conceptual section across the corridor from west to east; widths are indicative and imply no cross-section, setback or engineering conclusion



Core metrics (computed from the geometry)

Displayed values match metrics.json exactly



Protocol roles of the three areas and two wings

- Zhongzhiyuan AI Innovation Acceleration Area**
Protocol R&D and compute base | sense (hook) + knowledge (def) | 192.9 ha (provisional)
- Beijing AI Origin Community**
Protocol testbed with real residents | human adjudication + process (recipe) | 104.3 ha (provisional)
- Dazhongsi AI Industry Cluster**
Protocol commercialisation and showcase | process (recipe) + knowledge (def) | 72.0 ha (provisional)
- Zhongguancun service wing**
The protocol's service layer: government services packaged as standard APIs
- Xiaoyuehe scenario wing**
A live testbed for scenarios: one public experience route threading them all

Four-layer city OS

- Sense layer (hook)**
Urban events trigger governance automatically instead of stacking manual approvals
- Knowledge layer (def)**
Planning material and standards become queryable knowledge domains; AI and citizens draw from one source
- Process layer (recipe)**
Participation, piloting, evaluation and approval are standardised so anyone can copy and review them
- Human adjudication**
Every mechanism has a human adjudication gate; machines only advise and execute
- Spatial counterpart**
One protocol bus + three protocol loops + eight protocol nodes + one transverse section

Submitted boundary area
11.413 sq kmLand-use coverage
100.00 %Green ratio (self-computed)
19.34 %Public space ratio
6.17 %Key areas total
369.3 haAI scenario pilot unit
12 sites



Land Use Structure

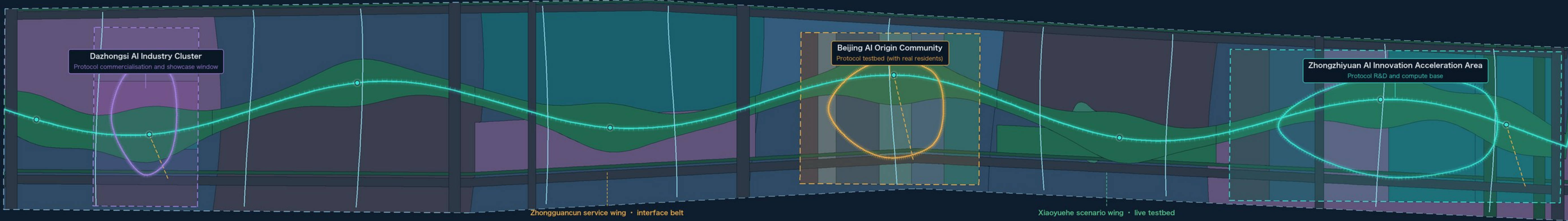
Land use is cut piece by piece from the submitted boundary: gap-free, overlap-free, fully tiled; adjacent polygons share vertices, union(land_use) = site_boundary

FIG 02 / LAND USE

PROVISIONAL · not an official redline

Land use (EPSG:4326 exchange / EPSG:4548 areas)

Colours grouped by land-use family: teal = research and services · deep green = green and open space · grey-blue = housing and fabric · grey = roads and reserve



Land-use composition

Areas computed from geometry/land_use.geojson in EPSG:4548; shares use the submitted boundary as denominator



Reserved land (16) accounts for 16.3%. This is a deliberate "protocol reserve": use and intensity are not fixed in this round but left for citizens and professional teams to decide together under the protocol — the direct land-use counterpart of "humans keep the final decision".

Structural logic: why this is not conventional colour-block zoning

One protocol = a bus along the length, a section across the width, a loop within each key area

Along: one protocol bus

The Jing-Zhang heritage alignment is abstracted into a continuous green spine with about 29.6 km of unbroken slow mobility, threading the three areas and two wings onto one bus so that an event on any segment travels to the rest.

Across: four-layer protocol section

From west to east: buffer green, knowledge-layer fabric, process service belt, sense-layer road corridor. The layering is not formal decoration — it places different governance functions where they can see one another.

Local: three closed loops

Each key area closes into a slow-mobility loop with the four layers arranged along it, so that "propose — pilot — evaluate — decide — capture" can run repeatedly at walking scale.

Reserve: clauses not yet decided

186 ha of reserved land sits across the wings and the fabric, standing for clauses of the governance protocol not yet voted on — so a one-off blueprint does not lock in the future.

Land-use legend and topology self-check

Land-use codes (project subset of the national Land and Sea Use Classification Guideline)



● gap between union(land_use) and the boundary

0.47 sqm

● Largest pairwise overlap between land-use polygons

0.02 sqm

● union(land_use) outside the boundary

2.96 sqm

● Threshold (set by this proposal, stricter than the repository's 1141 sqm)

11.5 sqm

Sources: Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources announcement (DATA-SRC-OFFICIAL-ANNOUNCEMENT20260509) + repository provisional boundaries provisional_boundaries.geojson (DATA-SRC-PROVISIONAL-BOUNDARIES20260605); spatial features and metrics recomputed from this proposal's geometry/"geojson in EPSG:4548.

Land-use designations are conceptual suggestions. They do not constitute land-use approval, tenure judgement, change of land-use classification, or retain/revise/demolish conclusions. Existing fabric features carry geometry, row-existing, condition and confidence/low. Road corridors are located roughly from public road names, used only to express structural relationships. They imply no road redline, alignment or cross-section conclusion.



Key Areas

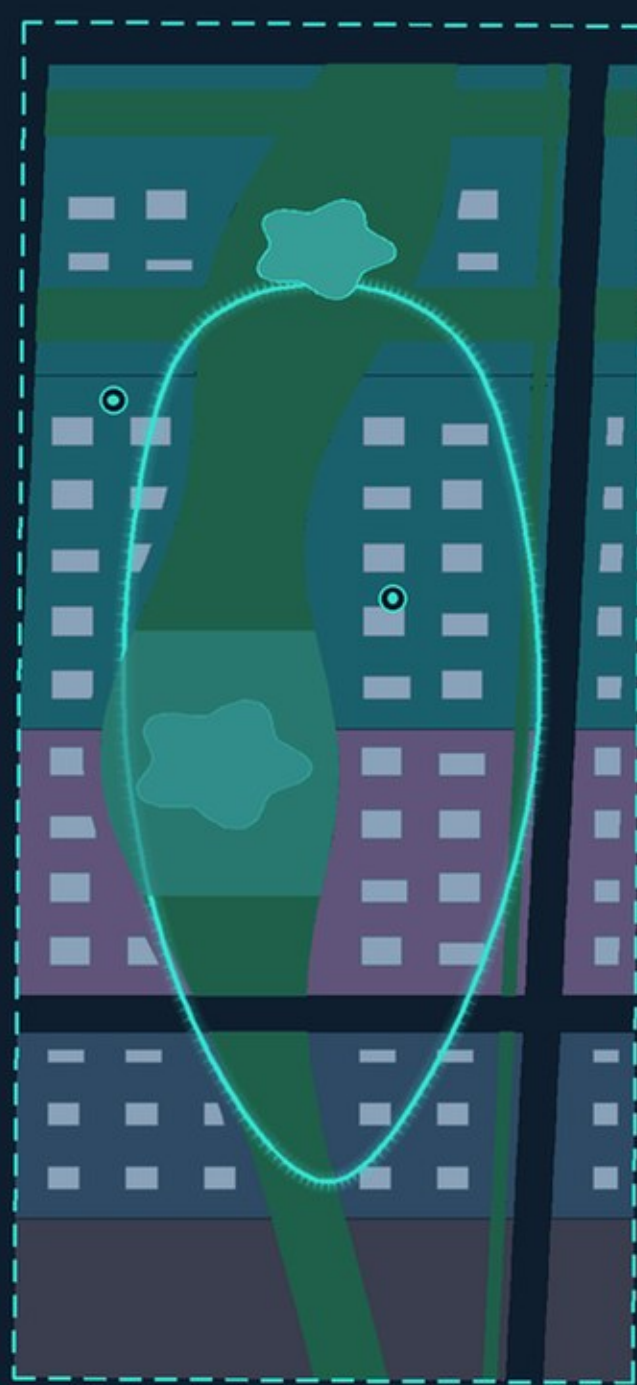
One four-layer protocol landing differently in each key area: R&D base / testbed / showcase window, each closing into its own loop

FIG 03 / KEY AREAS

PROVISIONAL · not an official redline

Zhongzhiyuan AI Innovation Acceleration Area

Protocol R&D and compute base | sense (hook) + knowledge (def)



0 300 m

Key area (provisional)
192.9 ha
announced about 192.1 ha, deviation +0.43%Protocol loop length / conceptual massing
3.24 km
The loop threads 63 conceptual massing footprints (no height or intensity conclusions)

How the four layers land here

Sense layer (hook) 17.7 ha

Compute and sensing facility (hook)

Knowledge layer (def) 32.6 ha

Model and knowledge-base R&D

Process layer (recipe) 23.2 ha

Protocol process and mastery services

Human adjudication 39.4 ha

R&D talent housing · protocol reserve unit

Beijing AI Origin Community

Protocol testbed (with real residents) | human adjudication + process (recipe)



0 300 m

Key area (provisional)
104.3 ha
announced about 104.3 ha, deviation +0.02%Protocol loop length / conceptual massing
2.08 km
The loop threads 43 conceptual massing footprints (no height or intensity conclusions)

How the four layers land here

Sense layer (hook) 15.7 ha

Research testing and pilot R&D

Knowledge layer (def) 12.0 ha

Education and developer training

Process layer (recipe) 10.1 ha

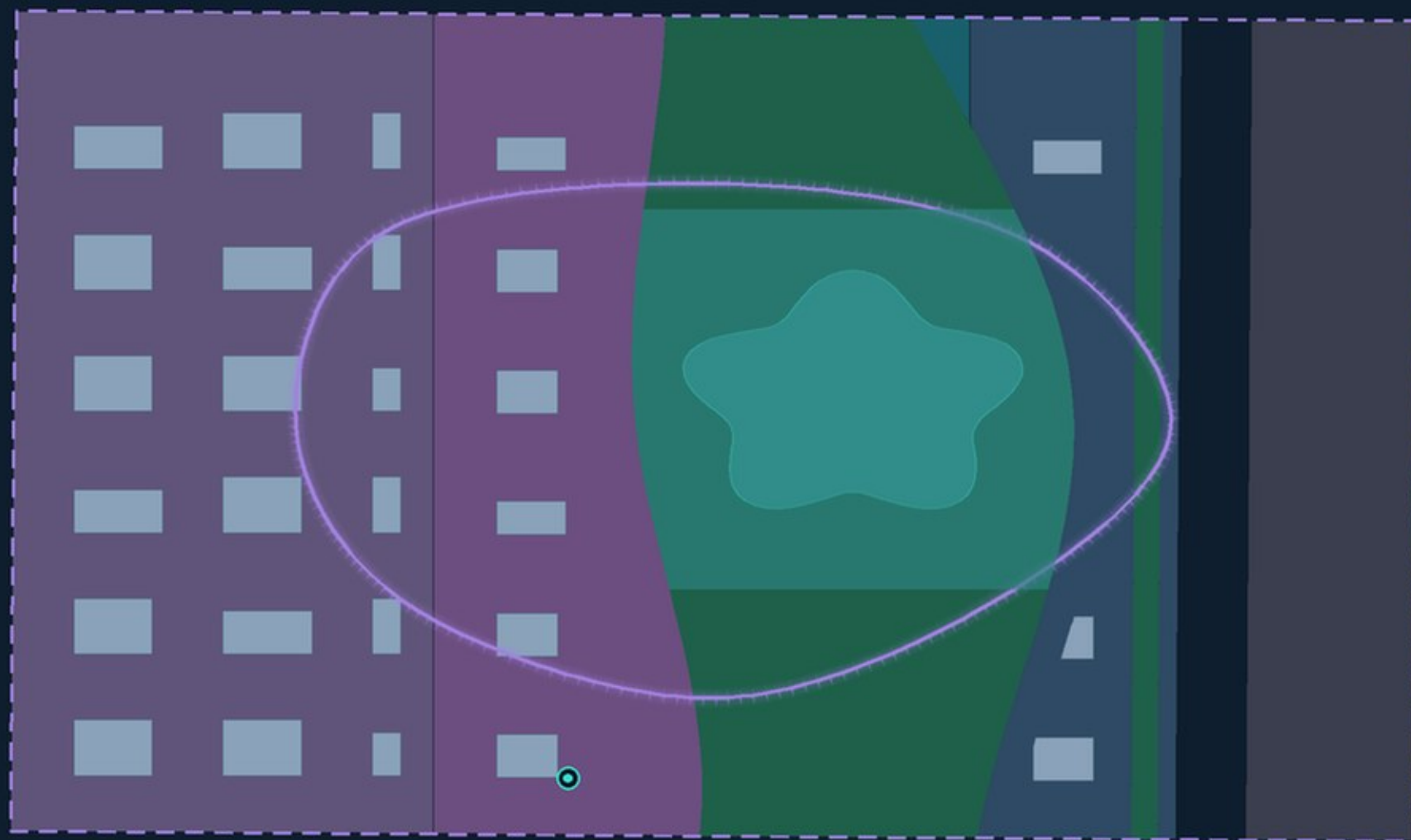
Community service and protocol-delegation facilities

Human adjudication 32.3 ha

Mixed resident and talent housing · community health and elder-care services · resident co-decision reserve unit

Dazhongsi AI Industry Cluster

Protocol commercialisation and showcase | process (recipe) + knowledge (def)



0 300 m

Key area (provisional)
72.0 ha
announced about 72.0 ha, deviation +0.00%Protocol loop length / conceptual massing
1.76 km
The loop threads 27 conceptual massing footprints (no height or intensity conclusions)

How the four layers land here

Process layer (recipe) 21.6 ha

Protocol commercialisation and trading services

Knowledge layer (def) 11.6 ha

Cultural display and protocol museum

Sense layer (hook) 0.2 ha

Industrial innovation and pilot-scale R&D

Human adjudication 15.3 ha

Station-integrated housing · forecourt protocol reserve unit

Comparison across the three areas

One protocol, different roles: who proposes, who pilots, who showcases

Key area	Protocol role	Layers carried	Dominant land use	AI scenario anchors	Human review gate
Zhongzhiyuan AI Innovation Acceleration Area	Protocol R&D and compute base	Sense (hook) + knowledge (def)	Research-led, with R&D talent housing and reserve	Low-speed delivery and inspection · energy and carbon ledger · open-data access point	Expert and authority review before each protocol release point
Beijing AI Origin Community	Protocol testbed (with real residents)	Human adjudication + process (recipe)	Housing + community services + education + health, with resident co-decision reserve	Rail correction and slow-mobility gaps · health service navigation · continuous accessible routes	Resident deliberation with a community-representative veto
Dazhongsi AI Industry Cluster	Protocol commercialisation and showcase window	Process (recipe) + knowledge (def)	Commercial services + cultural display + pilot-scale R&D	Cultural guide · civic proposal and review desk · enterprise service copilot	Compliance review for trading and operations; fact-checking of displayed content

Legend

- Conceptual massing footprint (no height or intensity)
- Protocol node plaza
- Protocol living room (heritage park segment)
- Green and open space
- Protocol loop (closed slow-mobility ring)
- AI scenario pilot unit

Sources: Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources announcement (DATA-SRC-OFFICIAL-ANNOUNCEMENT20260509) + repository provisional boundaries provisional_boundaries.geojson (DATA-SRC-PROVISIONAL-BOUNDARIES20260605); spatial features and metrics recomputed from this proposal's geometry/"geojson in EPSG:4548.

The rectangular key-area boundaries are inherited from provisional polygons inferred from the official announcement. Their edges must not be read as parcel redlines, road redlines or tenure boundaries. Internal zoning is a conceptual suggestion.

All AI scenarios are labelled "pending human review", "conceptual illustration", "not approved for operation". The authoritative scenario cards are in proposal.

Land use and key areas at a glance

- Gap-free, overlap-free, fully tiled

- Three key areas cut piece by piece from the submitted boundary, 100% coverage

Metrics: Zhongzhiyuan 192.9 ha / AI Origin Community 104.3 ha / Dazhongsi 72.0 ha (all provisional boundaries)

- Reserved land

- Boundary discipline, decided jointly by citizens and professional teams under the protocol

Key-area rectangle edges must not be read as parcel redlines, road redlines or tenure boundaries



Mobility & Blue-Green

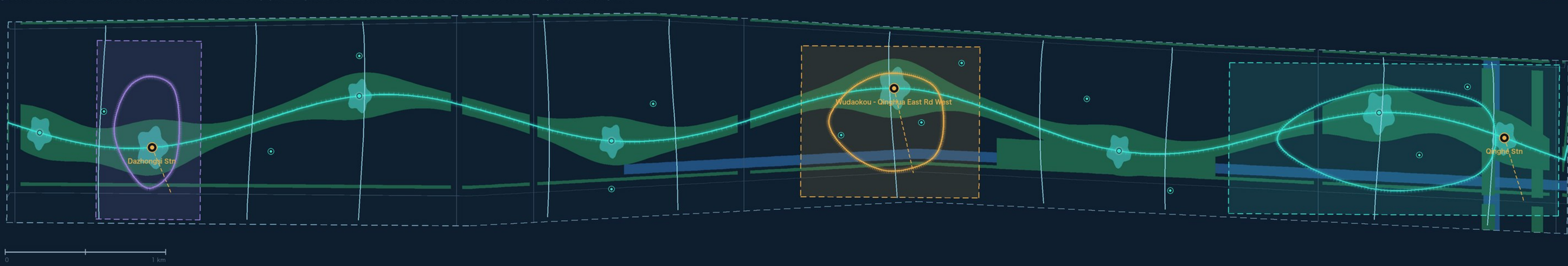
North - south continuity (one slow-mobility bus) × east - west stitching (ten connectors): severance handled in the language of loops and interfaces, with no bridge, tunnel or alignment conclusions

FIG 04 / MOBILITY & BLUE-GREEN

PROVISIONAL · not an official redline

Slow-mobility network and blue-green system

Existing road corridors are low contrast (indicative positions); the proposed slow-mobility system is high contrast. Water bodies and blue lines are indicative — the water authority's published data governs



Continuity in section: accessibility, slow mobility and blue-green all run unbroken

Widths and level changes are conceptual and imply no section design, setback or engineering feasibility conclusion

Continuous pedestrian and accessible frontage

Open ground-floor frontage runs continuously along the loop, ramps and tactile paving are not broken by site entrances

Cycling and low-speed shuttle

The loop doubles as a pilot route for low-speed delivery and inspection robots, with a human takeover gate

Blue-green continuity

Green spine + waterfront scenario belt + buffer greens form one unbroken ecological corridor

Protocol node access

The eight node plazas are shared access points for mobility, scenarios, deliberation and review

Mobility and blue-green metrics

All computed from the geometry, matching metrics.json

Proposed slow-mobility network length

29.58 km

Includes the spine bus, three protocol loops, ten east - west connectors and three rail connectors

Green ratio (self-computed)

19.3 %

220.7 ha of green and open space, not a statutory greenratio control value

Road land share

12.7 %

144.5 ha of road land (indicative corridor basis)

Existing road corridor length (indicative)

36.75 km

Plotted roughly from public road names; not a road redline or alignment

Public space ratio

6.2

70.5 ha of public space, measured as a union so no area is double counted

AI scenario pilot unit

12

6 anchored to the repository's official scenario registry, 6 added by this proposal, all pending human review

East - west stitching · north - south continuity

Interfaces rather than rhetoric: each stitching line is an access point that can be triggered by events and reviewed

CC01
1016 m
Pedestrian / accessibleCC02
1013 m
Pedestrian / accessibleCC03
1009 m
Pedestrian / accessibleCC04
1015 m
Pedestrian / accessibleCC05
1037 m
Pedestrian / accessibleCC06
1190 m
Pedestrian / accessibleCC07
1232 m
Pedestrian / accessibleCC08
1236 m
Pedestrian / accessibleCC09
1222 m
Pedestrian / accessibleCC10
1202 m
Pedestrian / accessible

North - south continuity: a roughly 9.7 km slow-mobility bus links the three areas and two wings into one chain, so an urban event anywhere on it propagates along the whole line. East - west stitching: ten pedestrian and barrier-free connectors cross today's severed interfaces and reattach the spine to the blocks on both sides. Together they form a 'bus + interface' urban topology — this proposal's protocol reading of 'stitch east - west, connect north - south', with no conclusions on bridges, tunnels, underground space or road alignment.

Legend
— Slow-mobility bus (spine)
— East - west stitching connectors-- Rail station connection (indicative)
— Watercourse blue-line control belt (indicative)Green and open space
— Existing road corridor (indicative)

Sources: Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources announcement (DATA-SRC-OFFICIAL-ANNOUNCEMENT20200509) + repository provisional boundaries provisional_boundaries.geojson (DATA-SRC-PROVISIONAL-BOUNDARIES20200605); spatial features and metrics recomputed from this proposal's geometry/"geojson in EPSG:4548.

Rail station positions are rough indications drawn from clues in the announced brief. They do not represent rail alignments, station positions or engineering conclusions. This proposal gives no conclusions on road alignment, bridges, tunnels or underground feasibility. Robotics and low-speed shuttle scenarios are all labelled "pending human review", "conceptual illustration", "not approved for operation", with a human takeover gate.



Metrics & Evidence

Metrics are computed, not written: geometry → metrics.json → drawings / HTML / proposal, and every link can be recomputed independently

FIG 05 / METRICS & EVIDENCE

PROVISIONAL · not an official redline

Core metrics (status = known, all recomputable from the geometry)

Submitted boundary area
11.413 sq km

announced about 11.4 sq km, deviation +0.11%

Land-use coverage
100.00 %

unlanded_use = site_boundary

Green ratio (self-computed)
19.34 %

220.7 ha

Public space ratio
6.17 %

70.5 ha

Road land share
12.66 %

144.5 ha (indicative basis)

Key areas total
369.3 ha

announced about 368.4 ha, deviation +0.24%

Evidence chain: every link a number passes through

Every link can be verified independently by scripts/spatial_review.py and scripts/visual_review.py

1 Official announcement

Areas, written extents, key-area names and order
DATA-SRC-OFFICIAL-ANNOUNCEMENT-20200509

2 Provisional boundary

provisional_boundaries.geojson (official_boundary=false)
DATA-SRC-PROVISIONAL-BOUNDARIES-20200605

3 Proposal geometry

geometry/"geojson" - 9 layers - 268 features
Generated by this proposal; every feature carries source_type / confidence / geometry_role

4 Area recomputation

Areas and lengths computed after projecting to EPSG:4548
area_unit_declared = computed geometry, max deviation = 0.002 sqm

5 Metric registration

metrics.json - 51 metrics - known / unknown stated explicitly
Unknown state is messy; no guessed values fill the gaps

6 Public display

Drawings / report / visual/index.html
HTML declares data-metric = data-value so machines can compare

Explicit unknowns: nothing invented, no guessed values

Missing statutory conditions plus this proposal's own discipline together require these metrics to stay empty

total_floor_area_sqm status = unknown · value = null

Statutory storey, height and floor-area-ratio controls are absent from the public package. Under this proposal's discipline no development intensity conclusion is given, so total floor area is not inferred.

floor_area_ratio status = unknown · value = null

Statutory floor-area-ratio conditions are unreleased. FAR is a statutory intensity control, and this proposal gives no conclusion on it as a matter of discipline.

building_density status = unknown · value = null

Statutory building-density conditions are unreleased. The footprints in the geometry cover only the three key areas and are conceptual, so they cannot constitute a site-wide density control value.

building_height_m status = unknown · value = null

Official height controls, aviation and landscape view corridors, and heritage height limits are all unreleased. This proposal gives no height conclusion.

Two registers side by side: scenario cards ≠ spatial anchors

Both sides happening to hold 12 entries is a coincidence, not a correspondence: governance objects and spatial carriers are separate registers and must each be checked against their own source file

Governance objects: scenario cards

scenario card table in proposal.md, 12 in total

S-01 Slow-mobility gap diagnosis PUB-015
S-03 Open-source release collaboration no spatial anchor
S-05 Enterprise service cockpit PUB-017
S-07 Cultural guide and narrative PUB-019
S-09 Low-carbon compute scheduling PUB-021
S-11 Low-speed robot delivery pilot PUB-016S-02 Public space crowding alert PUB-020
S-04 Safety evaluation sandbox no spatial anchor
S-06 Elder-friendly and accessible travel PUB-022
S-08 Data compliance salon PUB-024
S-10 Event safety operations review PUB-026 / PUB-025
S-12 Health service navigation PUB-029

Spatial carriers: AI scenario pilot units

geometry/public_space.geojson, 12 in total

PUB-015 Rail connection and slow-mobility gaps S-01
PUB-017 Enterprise service cockpit S-05
PUB-019 Jing-Zhang heritage cultural guide S-07
PUB-021 Campus energy and carbon ledger S-09
PUB-023 Civic proposal and review desk no matching card
PUB-025 Annual event operations S-10PUB-016 Low-speed delivery and inspection S-11
PUB-018 Public safety operations review S-10
PUB-020 Community health service navigation S-12
PUB-022 Continuous accessible route S-08
PUB-024 Open data and knowledge-domain access point S-06
PUB-026 Street environment sensing S-02

What the relationship actually is: 10 cards have a spatial anchor; 2 of them (open-source release collaboration / safety evaluation sandbox) run on existing buildings and institutions and take no anchor of their own. 11 anchors map to a card; 1 (the civic proposal and review desk) anchors the propose - review - merge - rollback governance loop rather than any single card. One card, event safety operations review, lands on two anchors at once. The equal counts are therefore a coincidence, and the metric

Geometry self-check

Recomputed by an independent script reading only the written GeoJSON, never build-time objects in memory

● GeoJSON root type and required properties	268 / 268 passed
● Geometry validity in EPSG:4548	0 invalid geometries
● land_use fully tiles the boundary	gap 0.47 sqm (threshold 11.5)
● land_use has no overlaps	largest pairwise intersection 0.02 sqm (threshold 11.0)
● Adjacent polygons share vertices	88 adjacent pairs share vertices
● key_areas disjoint and inside the boundary	overlap 0.00 sqm / 0 outside
● phasing tiles without overlap	overlap 0.01 sqm / uncovered 0.01 sqm
● area_sqm declared matches computed area	max deviation 0.0005 sqm
● metrics.json matches the geometry	7 / 7 match
○ Official redline	Not yet released; all boundaries provisional

The full self-check report is in geometry-selfcheck.md. The repository's scripts/spatial_review.py returns PASS with only three provisional notices (KEY, AREA, PROVISIONAL), which stem from the organiser's data gap and do not block content scoring.

Spatial evidence for the six agent tasks

Every task maps to features in the geometry, not just to prose

agent.1 Overall concept and coordination

site_boundary / key_areas / land_use / phasing
Four-layer protocol section + spine + beta loops + eight nodes

agent.3 Scenario enablement and a lively city

public_space (AI_SERVICE_ZONE +12)
6 anchored to the official registry + 6 added by this proposal

agent.5 Cultural fusion narrative

constraints (heritage belt - advisory) + land_use 0803
Rail heritage retention belt + cultural display land

agent.2 Full-stack innovation system and ecosystem

land_use (0802 research + 05 services)
Zhongguancun service wing interface belt + Zhongyuan R&D base

agent.4 Public space and landmarks

public_space (living roads 3 + node places 8)
The heritage park as urban living room; nodes as protocol milestones

agent.6 Event programme and long-term operation

phasing (release / pilot / extension)
Protocol version management release → pilot → extend → revise

Data gaps and discipline

- Official redline
- All scenarios: all boundaries are treated as provisional and the whole package must be recomputed once it is
- Statutory control indicators
- Two registers: scenario cards and spatial anchors are separate registers; their equal counts are a coincidence, not a correspondence
- Roads and rail
- Citation discipline: roughly from public road names; no alignment, route, bridge, tunnel or underground
- Intervened company lists, investment figures, output values or policy commitments; every feature carries a source id and type